School of Technology, Design & Computer Application
College of Technology
Department of Computer Engineering

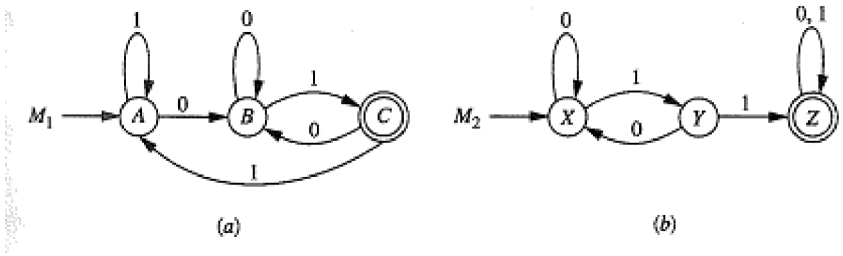
Semester:	VI	**Academic Year:**	2025-26
Course Name:	Computing Architecture and Computation	**Course Code:**	1010043426

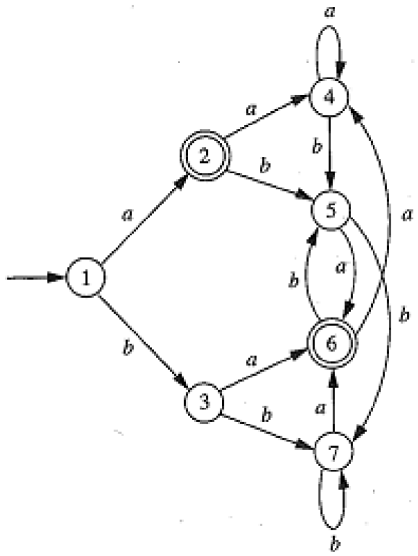
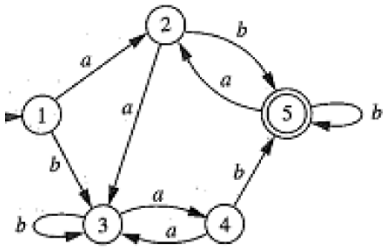
Question bank

Sr. No	Question	Marks	CO
	Unit : 1		
1	Define what a programming language grammar is and why it is important in language design.	4	CO1
2.	Explain the difference between context-free grammars and regular grammars in the context of programming languages.	4	CO1
3	Provide examples of tokens, terminals, and non-terminals in a programming language grammar.	4	CO1
4.	Describe the different classes of grammars based on the Chomsky hierarchy and their relevance in programming language theory.	7	CO1
5	Discuss the characteristics of regular, context-free, context-sensitive, and recursively enumerable grammars.	7	CO1
6	What is ambiguity in the context of grammatical specification?	4	CO1
7	Define scanning and parsing in the context of language processing.	4	CO1
8	Explain the difference between lexical analysis and syntax analysis.	4	CO1
9	Explain the concept of top-down parsing and its importance in language processing.	4	CO1
10	Compare and contrast top-down parsing with bottom-up parsing.	7	CO1
11	Describe the process of bottom-up parsing and its applications in language processing.	4	CO1
12	What are language processor development tools, and why are they important	4	CO1

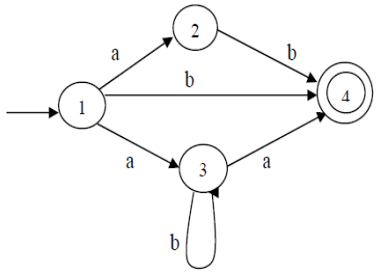
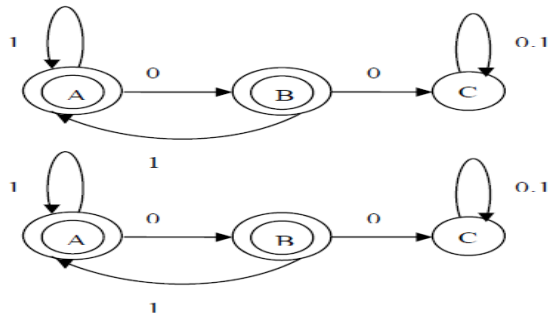
	in software development?		
13	Compare and contrast lexical analysis tools such as LEX with parser generators like YACC	4	CO1
14	Provide an overview of LEX and its purpose in language processing.	4	CO1
Unit : 2			
1	Explain the elements of assembly language programming and their significance.	4	CO2
2	Discuss the design considerations involved in developing an assembler.	4	CO2
3	Explain two pass design of assembler	7	CO2
4	Differentiate between various types of assemblers such as one-pass assemblers, two-pass assemblers, and multipass assemblers.	7	CO2
5	What are the causes of the large semantic gap between high-level programming languages and machine languages? How does it affect compiler design?	4	CO2
6	Define binding and binding times in the context of compiling. Provide examples of each.	4	CO2
7	Describe the data structures commonly used in compiling processes and their role in efficient code generation.	4	CO2
8	Explain scope rules in programming languages and how they influence compilation.	4	CO2
9	Discuss memory allocation strategies employed by compilers.	4	CO2
10	Outline the compilation process for expressions and control structures.	4	CO2
11	What is code optimization in compiling? Discuss various techniques used for code optimization.	4	CO2
12	Provide an overview of interpretation and how it differs from compilation.	4	CO2
13	Explain the procedures involved in debugging programs using interpreters and debuggers.	4	CO2
14	Classify debuggers based on their functionalities and usage scenarios.	4	CO2
15	Define macro definition and call in the context of macro processors.	4	CO2
16	Explain the process of macro expansion and its significance in code	4	CO2

	generation.		
17	Discuss nested macro calls and their handling in macro processing.	4	CO2
18	What are the functions of a macro processor in software development?	4	CO2
19	Identify and discuss the design issues commonly encountered in macro processors.	4	CO2
20	Introduce the concepts of linkers and loaders in the context of software development.	4	CO2
21	Describe the design of a linker and its role in generating executable code.	7	CO2
22	Explain dynamic linking and its advantages in software development.	4	CO2
23	Discuss different loading schemes employed by loaders.	7	CO2
24	Compare and contrast the functionalities of linkers and loaders, highlighting their respective roles in the compilation and execution process.	7	CO2
	Unit : 3		
1	Using Principle of Mathematical Induction, prove that for every n >= 1, $N \Sigma i = n (n+1) / 2$ i=0	4	CO3
2	Using Principle of Mathematical Induction, prove that for every n >= 1, $7 + 13 + 19 + . . . + (6n + 1) = n(3n +4)$	4	CO3
3	Explain Equivalence Relation.	4	CO3
4	Prove that √2 (square root of 2) is Irrational by method of Contradiction.	4	CO3
5	Using Principle of Mathematical Induction, Prove that For every n >= 1, $N \Sigma i^2 = n (n+1)(2n+1)/ 6$ i=1	4	CO3
6	Answer the following: (i) Give recursive definition of each of the following sets. a. The set T of positive integer divisible by 2 or 7. b. The set U of all string in {0, 1}* containing the substring 00. (ii) Prove that for any every n>=0, n (n^2+5) is divisible by 6.	7	CO3
7	Answer the following: (i) Given the relation R in A as R= {(1, 1), (2, 2), (2, 3), (3, 2), (4, 2), (4, 4)} is R (a) reflexive (b) symmetric (c) transitive? (d) Antisymmetric? (ii) Show that $2^n > n^3$ for n >10 by Mathematical Induction.	7	CO3

8	Define relation. Define reflexive and transitive relation. A binary relation R on NxN is defined as (a,b)R(c,d) if $a \leq c$ or $b \leq d$. Prove that R is reflexive but not transitive.	7	CO3
9	Define mathematical induction. Prove that if $0 < a < 1$ then $(1-a)^n \geq 1 - na$.	4	CO3
10	Using proof by contradiction, prove $\sqrt{3}$ is Not a rational number.	4	CO3
11	Define Mathematical Induction principle and prove that for any n>=4, $n!>2^n$	4	CO3
12	Explain following i. One to one and onto function ii. Properties of equivalence relation.	7	CO3
13	Use the principle of mathematical induction to prove that $1 + 3 + 5 + \ldots + r = n^2$ for all n>0 where r is an odd integer & n is the number of terms in the sum. (Note : r= 2n-1)	4	CO3
14	Let A = {1, 2, 3, 4, 5, 6} and R be a relation on A such that aRb iff a is a multiple of b. Write R. Check if the relation is i) Reflexive ii) Symmetric iii) Asymmetric iv) Transitive	7	CO3
15	Answer the following 1. Define regular language and regular expressions. 2. Find regular expression for the following: Language of all string that do not end with 01. 3. Describe the language corresponding to following: (1+01)*(0+01)*	4	CO3
16	Answer the following 1.Draw FA for regular expression: (111+100)*0 2. Let M1 and M2 be the FA in fig below for the language L1 and L2, find L1 U L2 and L1 ∩ L2.  	7	CO3
17	Write theorem: For any NFA M =(Q,Σ,q0,A,δ) accepting a language L, there is an FA M1 =(Q,Σ,q1,A1,δ1) that also accepts L.	7	CO3
18	Write Kleene's Theorem part-I, Any regular language can be accepted by a finite automation.	7	CO3

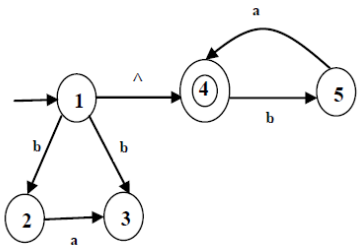
19	Answer the following 1. For following NFA find minimum FA accepting same language 	7	CO3
20	Answer the following 1. For following NFA find minimum FA accepting same language 	7	CO3
21	Write Regular Expressions for the following languages of all strings in {0,1}* (i) Strings that do not end with 01. (ii) Strings with odd numbers of 1's (Ones).	4	CO3
22	Define Nondeterministic Finite Automata (NFA) and write down recursive definition of δ* for NFA- Λ.	7	CO3
23	Write definition of Finite Automata and draw FA for the strings: (i) The string with next to last symbol as 0. (ii) The string with number of 0s odd and number of 1s odd.	4	CO3
24	Convert following NFA- Λ to NFA and FA.	7	CO3

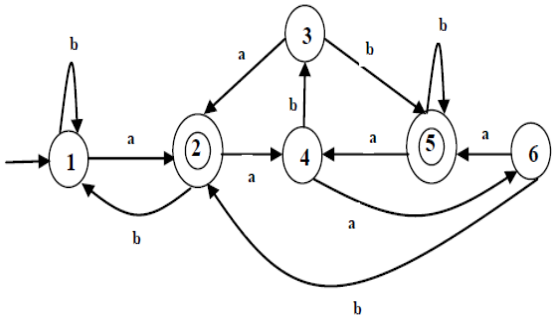
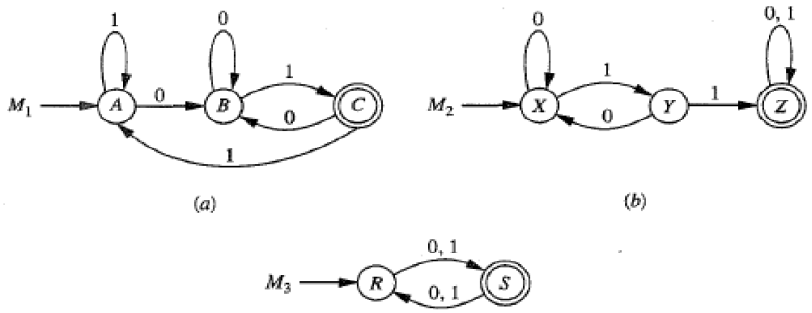
q	δ (q, Λ)	δ(q, 0)	δ(q, 1)
A	{B}	{A}	Ø
B	{D}	{C}	Ø
C	Ø	Ø	{B}

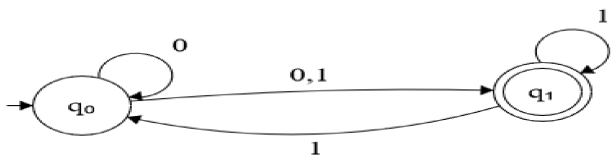
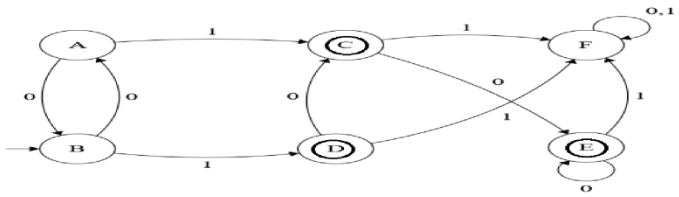
	D | Ø | {D} | Ø		
25	For the following Regular Expression draw an NFA- Λ recognizing the corresponding languages. (i) (00 + 1)* (10)* (ii) 001*0*11	7	CO3
26	Draw Finite Automata (FA) for following languages: L1 = {x / 00 is not a substring of x } L2 = {x / x ends with 01 } Find FA accepting languages (i) L1 ∩ L2 and (ii) L2 – L1	7	CO3
27	Compare FA, NFA and NFA- Λ with illustration.	7	CO3
28	Convert the following NFA into FA.  	7	CO3
29	Suppose that L1 and L2 are the subsets L_1 = L_2 =  Draw the FAs recognizing the following languages. • L1 ∩ L2 • L1 – L2	7	CO3
30	Define δ* for! FA- NFA and NFA-Λ. Also Calculate δ* (1, ab) and δ* (1, abaab) from the following transition table.	7	CO3

Q	$\delta(q, a)$	$\delta(q, b)$
1	{1, 2}	{1}
2	{3}	{3}
3	{4}	{4}
4	{5}	Ø
5	Ø	{5}

31	Attempt the following : • Draw FA for (11+110)* 0 • Write a Regular Expression for the String of 0's and 1's in which string ends with 1 and does not contain substring 00.	7	CO3
32	Languages L1 and L2 are the subsets given below. Where Σ = { 0 , 1 } L1 = { x | 00 is not a substring of x } L2 = { x | x ends with 01 } Draw FAs recognizing the following languages (i) L1 - L2 (ii) L1∩L2	7	CO3
33	Draw a DFA that recognize the language of all strings of 0's and 1's of length at least 1 that, if they were interpreted as binary representation of integers, would represent evenly divisible by 3. Your DFA should accept the string 0 but no other strings with leading 0's.	7	CO3
34	Enlist applications where the finite automaton is useful. Also Find a string of minimum length in {0,1}* not in the language corresponding to the regular expression : 1*(0+10)*1*	7	CO3
35	Explain the procedure for converting the given DFA in to minimum number of state DFA. Using this procedure convert the following DFA into minimum number of states DFA (minimized FA) where Σ = {0,1}. <table><tr><td>Q</td><td>δ(q,a)</td><td>δ(q,b)</td></tr><tr><td>- +1</td><td>{3}</td><td>{2}</td></tr><tr><td>2</td><td>{4}</td><td>{1}</td></tr><tr><td>3</td><td>{5}</td><td>{4}</td></tr><tr><td>4</td><td>{4}</td><td>{4}</td></tr><tr><td>5</td><td>{3}</td><td>{2}</td></tr></table>	7	CO3
36	Define Dead-End State with Example.	4	CO3
37	Define Pumping Lemma for Regular Languages. Prove that the language L = {a^n: n is a prime number} is not regular.	4	CO3
38	Check the validity of the following equality with proper reason. (0 0 * 1) * 1 = 1 + 0 (0 + 1 0) * 1 1	4	CO3
39	Write Regular Expressions for the following languages of all strings in {0,1}* (i) Strings that contains odd number of 0's (zeroes). (ii) Strings that begin or end with 00 or 11.	4	CO3
40	Write definition of finite automata and draw FA for the strings: (i)The string in {0,1}* ending in 10 or 11. (ii)The string corresponding to Regular expression {11}*{00}*	7	CO3

<table>
<tr><td>41</td><td>Convert following NFA- Λ to NFA and FA.
<table>
<tr><th>Q</th><th>δ(q, Λ)</th><th>δ(q, 0)</th><th>δ(q, 1)</th></tr>
<tr><td>A</td><td>{B,D}</td><td>{A}</td><td>Ø</td></tr>
<tr><td>B</td><td>Ø</td><td>{C}</td><td>{E}</td></tr>
<tr><td>C</td><td>Ø</td><td>Ø</td><td>{B}</td></tr>
<tr><td>D</td><td>Ø</td><td>{E}</td><td>{D}</td></tr>
<tr><td>E</td><td>Ø</td><td>Ø</td><td>Ø</td></tr>
</table></td><td>4</td><td>CO3</td></tr>
<tr><td>42</td><td>Compare FA, NFA and NFA- Λ. For the following Regular Expression draw an NFA-Λ recognizing the corresponding language. (0 + 1)* (10+01)* 11</td><td>7</td><td>CO3</td></tr>
<tr><td>43</td><td>Draw Finite Automata (FA) for following languages:
L1 = {x / 11 is not a substring of x, x Є {0,1}*}
L2 = {x / x ends with 10, x Є {0,1}* }
Find FA accepting languages (i) L1 ∩ L2 and (ii) L1 – L2</td><td>4</td><td>CO3</td></tr>
<tr><td>44</td><td>Use the pumping lemma to show that following language is not regular: L = {xy|x,y ϵ {0,1}* and y is either x or x^r}</td><td>4</td><td>CO3</td></tr>
<tr><td>45</td><td>Write Regular Expressions for the following languages of all strings in {0,1}*
(i) Strings that start with 1 and do not end with 10.
(ii) Strings with length 6 or less.</td><td>7</td><td>CO3</td></tr>
<tr><td>46</td><td>For the following Regular Expression draw an NFA- Λ recognizing the corresponding language.
(0 + 1)* (10+110)* 1</td><td>7</td><td>CO3</td></tr>
<tr><td>47</td><td>Draw FA for accepting:
(i)The string in {0,1}* ending in 1 and not containing substring 00.
(ii)The strings with odd no. of 1’s and odd no. of 0’s</td><td>4</td><td>CO3</td></tr>
<tr><td>48</td><td>Convert the following NFA- Λ into FA.
 
</td><td>7</td><td>CO3</td></tr>
<tr><td>49</td><td>Minimize the following DFA (If Possible).</td><td>7</td><td>CO3</td></tr>
</table>

			
50	Attempt the following : ● Draw FA for (a + b)* baaa. Write a Regular Expression for the String of 0's and 1's in which number of 0's and 1's are even.	7	CO3
51	Describe the language corresponding to following: (1+01)*(0+01)*	4	CO3
52	Find a regular expression corresponding to each of the following subsets of {0, 1}*. i. The language of all strings that do not contain the substring 110. ii. The language of all strings containing both 101 and 010 as substrings. iii. The language of all strings in which both the number of 0's and the number of l's are odd.	7	CO3
53	For each of the following regular expressions, draw an FA recognizing the corresponding language. i. 1(01 + 10)* + 0(11 + 10)* ii. (010 + 00)*(10)*	7	CO3
54	Let M1, M2 and M3 be the FAs pictured in Figure below, recognizing languages L1, L2, and L3 respectively.  Draw FAs recognizing the following languages. i. L1 U L2 ii. L1 ∩ L2 iii. L1 - L2 iv. L1 ∩ L3	7	CO3

	v. L3 - L2		
55	Define language. Draw Deterministic Finite Automata for the following languages i) L1 = { x ε (0,1)* | x contains 110111} ii) L2 = { x ε (0,1)* | x contains odd number of zero and even number of 1} iii) L3 = { x ε (0,1)* | x do not contains 110 }	7	CO3
56	Define NFA and NFA-Λ. Convert the following NFA to DFA  	7	CO3
57	Minimize the following FSM  	7	CO3
58	Write regular expression for the following languages i) L1 = {x ε (0,1)* | x do not ends with 11} ii) L2 = {x ε (0,1)* | x contains both 101 and 110}	4	CO3
59	Draw FA for each of the following RE. i. (0+1)*(1+00)(0+1)* ii. (0+1)*(01+110) iii. (111+100)*0	7	CO3
60	For each of the RE draw NFA-Λ i. (0+1)*(011+01010)(0+1)* ii.(0+1)(01)*(011)*	7	CO3
61	Find regular expression for following i. Language of all strings containing exactly two 0's. ii. Language of all strings that begins or ends with 00 or 11. iii. Language of all strings in which every 0 is followed immediately by 11.	7	CO3
62	Find minimum state FA recognizing the language corresponding to the following RE. i. (0*10+1*0)(01)* ii. (010)*1 + (1*0)*	7	CO3
63	Convert the NFA given in Table below to its corresponding DFA and draw the DFA.	7	CO3

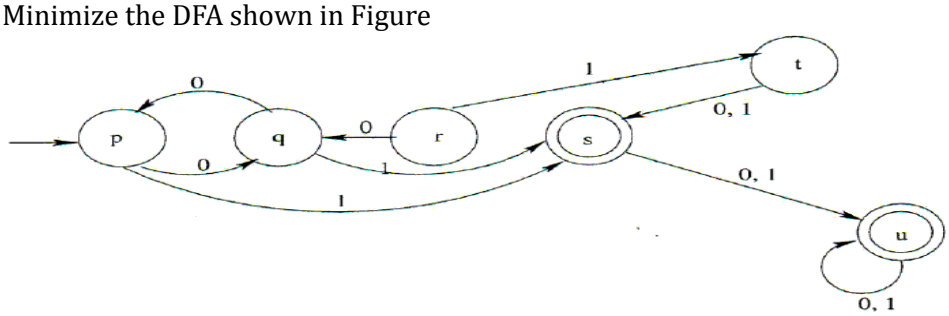
Current State	Input symbol	
	0	1
→Q_0	Q_1	Q_0, Q_2
Q_1	Q_2	Q_0
Q_2 *	Q0	---

64	If L = $\{0^i 1^i \mid i \geq 0\}$ Prove that L is regular.	4	CO3
65	Write a regular expression for language L over {0,1} such that every string in L i) Begins with 00 and ends with 11. ii) Contains alternate 0 and 1.	4	CO3
66	Give recursive definitions of the extended transition functions, δ̂(i.e., for strings) for DFA and NFA.	7	CO3
67	Consider the NFA-Λ depicted in following table:	7	CO3

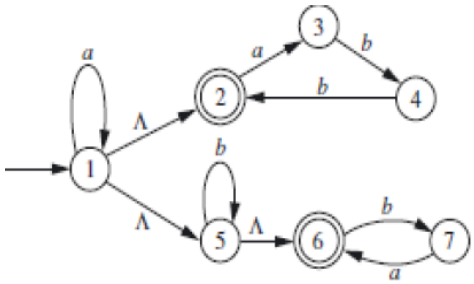
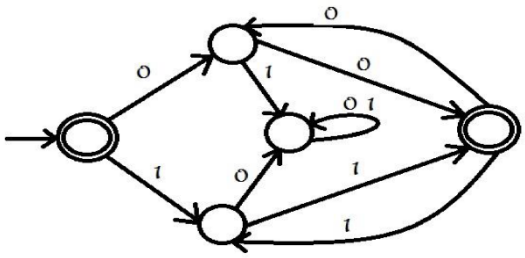
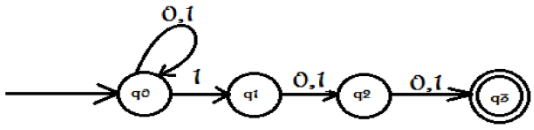
	Λ	a	b	c
→p	Φ	{p}	{q}	{r}
q	{p}	{q}	{r}	Φ
* r	{q}	{r}	Φ	{p}

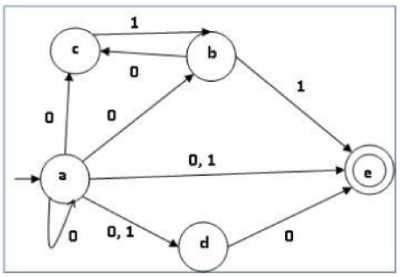
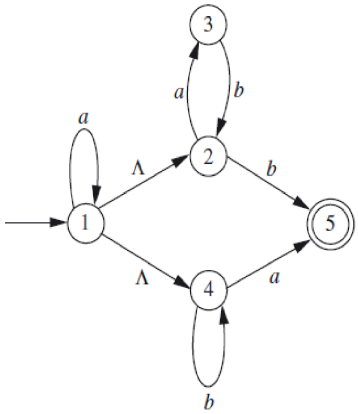
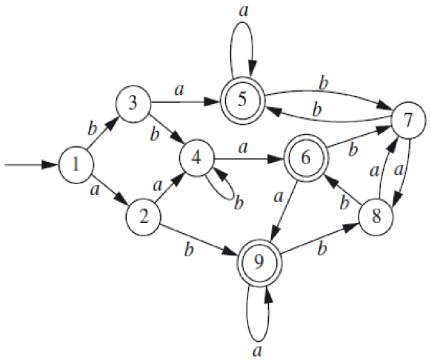
(i) Compute the Λ-closure of each state.
(ii) Convert the NFA-Λ to a DFA.

68	Minimize the DFA shown in Figure	7	CO3



69	Using constructive approach determine NFA- Λ for the regular expression (0 + 1)*1(0 + 1).	7	CO3
70	Define language. Draw Deterministic Finite Automata for the following languages i) L1 = { x ε (0,1)* | x contains101} ii) L2 = { x ε (0,1)* | x contains odd number of zero and ends with 00} iii) L3 = { x ε (0,1)* | x ends with 11}	7	CO3
71	Figure shows NFA-^. Draw an FA accepting the same language.	7	CO3

			
72	What is the regular expression of following FA? 	7	CO3
73	Draw NFA – Λ for ((0 + 1)*10 + (00)*(11)*)* Show step by step construction.	7	CO3
74	An NFA with states 1-5 and input alphabet {a, b} has the following transition table. q $\delta(q,a)$ $\delta(q,b)$ 1 {1,2} {1} 2 {3} {3} 3 {4} {4} 4 {5} Ø 5 Ø {5} *Q.1 Draw its transition diagram* *Q.2 Calculate δ^* (1,a)* *Q.3 Calculate δ^* (1,aaabaab)*	7	CO3
75	Convert this NFA to FA 	7	CO3
76	A language L {a, b}* is defined as follows: 1. a ϵ L 2. For any x ϵ L, ax ϵ L 3. For any x and y in L, all the strings bxy, xby and xyb are in L 4. No other strings are in L. Prove that every element of L has more a's than b's.	7	CO3
77	Convert the following NDFA to DFA.	7	CO3

			
78	Write Regular Expression over the alphabets {a, b} consisting strings: ● Second last character as 'a' Starting with 'a' and ending with 'b'	4	CO3
80	Construct fin finite automata for following left linear grammar: S → X0 | Y1 X → Y1 Y → Y0 | 1	7	CO3
81	Convert the following NFA - Λ into its equivalent DFA that accepts the same language: 	7	CO3
82	Find a minimum-state FA for the following FA that recognizes the same language using the minimization algorithm: 	7	CO3

83	Define DFA and NFA and NFA- Λ	4	CO3
	Unit : 4 & 5		
1	Write theorem: If L1 and L2 are context free languages, then the language L1 U L2, L1∩L2 and L1* are also CFLs.	4	CO4
2	Answer the following 1.Find context free grammar generating following language{$a^ib^jc^k$ | i = j or i = k} 2. Show that CFG S→a|Sa|bSS|SSb|SbS is ambiguous. 3. find an equivalent unambiguous grammar for following: S →A|B A→aAb|ab B→abB|Λ	7	CO4
3	Explain bottom up parsing with examples.	4	CO4
4	Define CFG. Prove that the following CFG is Ambiguous. S → S + S | S * S | (S) | a Write the unambiguous CFG for the above grammar.	7	CO4
5	Answer the following. (i) Design a CFG for the following language. L = { $0^i1^j0^k$ / j > i + k } (ii) Give the difference between Top Down Parsing And Bottom Up Parsing.	7	CO4
6	Prove: There are context-free languages L1 and L2 so that L1 ∩ L2 is not a CFL and there is a CFL L so that L' is not a CFL	4	CO4
7	Given the CFG G, find a CFG G' in Chomsky Normal form generating L(G) – { Λ} S→ AaA | CA | BaB A→ aaBa | CDA | aa | DC B→ bB | bAB | bb | aS C→ Ca | bC | D D→ bD | Λ	7	CO4
8	Explain Derivation Tree, Expression Tree and Ambiguity with Example.	4	CO4
9	For the following CFG's, describe the language it accepts. 1. S → SS | XaXaX | ^ X → bX | ^ 2. S→aM | bS M→aF | bS F→aF | bF | ^ 3. S→aS | bS | a | b | ^	4	CO4
10	Find CFG for the following languages. 1. L = { $a^ib^jc^k$ | j > i + k } 2. L = { $a^ib^jc^k$ | i = j or j = k }	4	CO4

11	Given the context-free grammar G, find a CFG G' in Chomsky Normal Form generating L(G) – {^}. G has production S→S(S) | ^	4	CO4
12	Define Context Free Grammar (CFG). Describe the language accepted by following CFG: S→aSa | bSb | a | b | Λ	4	CO4
13	Prove that the following CFG is Ambiguous. S→S + S | S * S | (S) | a Write the unambiguous CFG for the above grammar. Draw Parse tree for the string a + a * a.	4	CO4
14	Convert the following CFG to the equivalent Chomsky Normal Form(CNF). S→AACD | ACD | AAC | CD | AC | C A→aAb | ab C→aC | a D→aDa | bDb | aa | bb	7	CO4
15	Show that CFG S→a|Sa|bSS|SSb|SbS is ambiguous.	4	CO4
16	find an equivalent unambiguous grammar for following: S→A|B A→aAb|ab B→abB|Λ	4	CO4
17	Let L be the language corresponding to the regular expression (011+1)* (01)*. Find the CFG generating L.	4	CO4
18	Given the CFG *G*, find a CFG *G'* in Chomsky Normal form generating L(G) – { Λ} *S→A | B | C* *A→aAa | B* *B→bB | bb* *C→aCaa | D* *D→baD | abD | aa*	7	CO4
19	Define CFG and Design a CFG for the following language. L = { x ∈ {0,1}* | $n_0(x) \neq n_1(x)$ }	4	CO4
20	Differentiate Regular Grammars and Context Sensitive Grammars.	4	CO4
21	Generate the Context-Free Grammars that give the following languages. (i) {w | w contains at least three 1s} (ii) {w | w starts and ends with the same symbol}	4	CO4
22	Define Context Free Grammar (CFG). Design CFG for Generating Following Language: (1) For Balanced Parenthesis (2) Set of even length strings in {a, b, c, d}* with two middle symbol equal.	4	CO4

<table>
<tr><td>23</td><td>Design an ambiguous grammar for if-then-else statement that also generates if-then statement. Re-write an equivalent unambiguous grammar. Prove that Grammar is Unambiguous by tracing “ic_1tic_2taea”.</td><td>4</td><td>CO4</td></tr>
<tr><td>24</td><td>Find CFG from given PDA that accepts the language {0n1n}. PDA is (Q, Σ, Γ, δ, q, Z, F) where Q={q, r}, Σ = {0, 1}, Γ = {Z, X}, δ is defined by:
<table>
<tr><th>State</th><th>Input</th><th>Stack</th><th>New State</th><th>Stack</th></tr>
<tr><td>q</td><td>0</td><td>Z</td><td>q</td><td>XZ</td></tr>
<tr><td>q</td><td>0</td><td>X</td><td>q</td><td>XX</td></tr>
<tr><td>q</td><td>1</td><td>X</td><td>r</td><td>^</td></tr>
<tr><td>r</td><td>1</td><td>X</td><td>r</td><td>^</td></tr>
<tr><td>r</td><td>^</td><td>Z</td><td>r</td><td>^</td></tr>
</table></td><td>7</td><td>CO4</td></tr>
<tr><td>25</td><td>(1) Given the Context Free Grammar G, find a CFG G’ in Chomsky Normal Form generating L(G) – { }
S → SS | A | B
A → SS | AS | a
B → /\
(2) Convert following CFG to PDA
S → 0S1 | 00 | 11</td><td>4</td><td>CO4</td></tr>
<tr><td>26</td><td>Given the Context Free Grammar G, find a CFG G’ in Chomsky Normal Form generating L(G) – { }
1) S → aY | Ybb | Y
X → /\ | a
Y → aXY | bb | XXa
2) S → AA
A → B | BB
B → abB | b | bb</td><td>7</td><td>CO4</td></tr>
<tr><td>27</td><td>Define Context Free Grammar. Design a CFG for the following language.
L = { x ε (0,1)* | $n_0(x) = n_1(x)$}</td><td>4</td><td>CO4</td></tr>
<tr><td>28</td><td>For the following CFG, Find Chomsky normal form
S->AACD A->aAb|ʌ C-> aC|a D->aDa|bDb|ʌ</td><td>4</td><td>CO4</td></tr>
<tr><td>29</td><td>Convert the CFG, G ({S,A,B},{a,b},P , S) to CNF , where P is as follows
S --> aAbB A --> Ab | b B --> Ba | a</td><td>7</td><td>CO4</td></tr>
<tr><td>30</td><td>Prove that the following CFG is Ambiguous.
S -> S + S | S * S | a | b
Write the unambiguous CFG based on precedence rules for the above grammar.
Derive the parse tree for expression (a + a)*b from the unambiguous grammar.</td><td>4</td><td>CO4</td></tr>
<tr><td>31</td><td>Given the Context Free Grammar G, find a CFG G’ in Chomsky Normal Form generating L(G) – { }</td><td>7</td><td>CO4</td></tr>
</table>

	S → aY | Ybb | Y X → /\ | a Y → aXY | bb | XXa		
32	Consider following grammar: S → A1B A → 0A | Λ B → 0B | 1B | Λ Give leftmost and rightmost derivations of the string 00101. Also draw the parse tree corresponding to this string.	7	CO4
33	Consider following grammar: S → ASB | Λ A → aAS | a B → SbS | A | bb i. Eliminate useless symbols, if any. ii. Eliminate Λ productions.	7	CO4
34	Prove that the following language is ambiguous and convert into unambiguous E → E + E | E * E | id	4	CO4
35	Convert the following language in Chomsky normal form. S→ASB | SAB A→ BC B→ bB | c C → e	7	CO4
36	(i) Write a CFG for solving simple (& parenthesized) expression, such as + and *. (ii) Also write CFG fir regular expression r = (a + b) (a + b + 0 + 1)* Use CFG defined for part (i). (iii) Derive the string (which is defined in part (ii)) a * (a + b00) by applying left most derivation and right most derivation.	7	CO4
37	Define Context Free Grammar. Design a CFG for the language L = { $a^i b^j c^k$ | i ≠ j + k }	4	CO4
38	Define Context Free Grammar. Find context-free grammar for the language: L = {$a^i b^j$ | i < 2j}	4	CO4
39	Explain Union Rule and Concatenation Rule for Context-Free Grammar.	4	CO4
40	Decide whether the given language is a CFL, and prove your answer. L = { xyx | x, y ϵ {a, b}* and |x| ≥ 1}	4	CO4
41	Define Context-Sensitive Grammar. What is the language of following context-sensitive grammar? S → aTb | ab aT → aaTb | ac.	4	CO4
42	What is CNF? Convert the following CFG into CNF.	4	CO4

	S → ASA | aB, A → B | S, B → b | ε		
43	Consider the grammar: S→ABA, A→aA | ε, B→bB | ε Is given grammar ambiguous? If so then remove ambiguity	4	CO4
44	Find context free grammar for the following language. L1={ $a^i b^j c^k$ | i = j + k}, L2= (011+1)* (01)*, L3=(0+1)1*(1+(01)*)	4	CO4
45	Give the left linear grammar for RE (10)*1	4	CO4
46	Eliminate useless symbols, ε -productions and unit productions for the following grammar: S→0A0 | 1B1 | BB, A→C, B→S | A, C→S | ε	4	CO4
47	Give CFG for following languages: 1). L = a*b* 2). L = {a^{n+2} b^n | n >= 0}	4	CO4
48	Prove that – "If there is a CFG for the language L that has no Λ-productions, then there is a CFG for L with no Λ-productions and no unit productions". Support your answer with the help of the following CFG: S →A | bb A →B | b B →S | a	4	CO4
49	Convert the following CFG into its equivalent CNF: S → TU | V T → aTb | Λ U→ cU | Λ V →aVc | W W →bW | Λ	4	CO4
50	Write transition table for PDA recognizing following language: { $a^i b^j c^k$ | j = i or j = k }.	7	CO5
51	For the language L = { xcx^r / x Є {a,b}* } design a PDA(Push Down Automata) and trace it for string "abcba".	7	CO5
52	Design and draw a deterministic PDA accepting "Balanced strings of Brackets" which are accepted by following CFG. S→ SS | [S] | { S } | Λ	7	CO5
53	Define PDA and design PDA for L = { x ∈ { a, b}* | $n_a(x) > n_b(x)$ }	7	CO5
54	Give transition table for deterministic PDA recognizing the following language. { a^n b^{n+m} a^m | n,m ≥ 0)	7	CO5
55	Give transition table for deterministic PDA recognizing the following language. { $a^i b^j c^k$ | i, j, k ≥ 0 and j = i or j = k }	7	CO5

56	Design and draw a deterministic PDA accepting strings with more a's than b's. Trace it for the string "abbabaa".	7	CO5
57	Prove: The language pal= { $x \in \{a, b\}^*$ | $x = x^r$ } cannot be accepted by any deterministic pushdown automaton.	7	CO5
58	Write PDA for following languages: { $x \in \{a,b,c\}^*$ | $n_a(x) < n_b(x)$ or $n_a(x) < n_c(x)$ }.	7	CO5
59	For the language L={set of strings over alphabet {a, b} with exactly twice as many a's as b's} design a PDA (Push Down Automata) and trace it for the string "abaabbaaaaabaab"	7	CO5
60	For the language L={ $a^i b^j c^k$ | i, j, k $\geq$ 0 and i + j = k } design a PDA (Push Down Automata) and trace it for String "bbbbbccccc"	7	CO5
61	Given a CFG , G =({S,A,B},{0,1},P,S) with P as follows S --> 0B| 1A A --> 0S|1AA|0 B --> 1S| 0BB | 1 Design a PDA M corresponding to CFG, G. Show that the string 0001101110 belongs to CFL , L(G)	7	CO5
62	Consider following PDA machine M = ({p, q}, {0,1}, (x, z}, δ, q, Z) where δ is given by δ(q, 1, z) = (q, xz) δ(q, 1, x) = (q, xx) δ(q, ^, x) = (q,^) δ(q, 0, x) = (p, x) δ(p, 1, x) = (p, ε) δ(p, 0, z) = (q, z) Construct Equivalent CFG.	7	CO5
63	For the PDA, ({q0, q1}, {0, 1}, {0, 1, z0}, δ, q0, z0, ϕ), where δ is δ(q0, ε, z0) = {(q1, ε)} δ(q0, 0, z0) = {(q0, 0z0)} δ(q0, 0, 0) = {(q0, 00)} δ(q0, 1, 0) = {(q0, 10)} δ(q0, 1, 1) = {(q0, 11)} δ(q0, 0, 1) = {(q1, ε)} δ(q1, 0, 1) = {(q1, ε)} δ(q1, 0, 0) = {(q1, ε)} δ(q1, ε, z0) = {(q1, ε)} Obtain CFG accepted by the above PDA.	7	CO5
64	Define PDA and give PDA to accept strings of palindrome. Show trace on the string baab	7	CO5
65	Design pushdown automata to check well-formed parenthesis.	7	CO5

66	Convert the following CFG into its equivalent PDA. $S \rightarrow AB$ $A \rightarrow BB$ $B \rightarrow AB$ $A \rightarrow a$ $B \rightarrow a \mid b$	7	CO5
67	Show using the pumping lemma that the following language is not a CFL. $L=\{a^i b^j c^k \mid i < j < k\}$	7	CO5
68	Explain push down automata with example.	7	CO5
69	Prove that the context-free languages are closed under union.	7	CO5
	Unit : 6		
1	Write TM accepting Palindrome.	7	CO5
2	Write TM accepting $\{ss \mid s \in \{a,b\}^*\}$	7	CO5
3	Define Turing Machine. Describe its capabilities. Also write short notes on Universal Turing Machine.	7	CO5
4	Draw a Turing Machine(TM) to accept Palindromes over {a,b}. (Even as well as Odd Palindromes)	7	CO5
5	Explain Universal TM and Church Turing Thesis	4	CO5
6	Explain Cook's Theorem.	4	CO5
7	Draw a transition diagram for a Turing machine accepting the following language. $\{ a^n bn c^n \mid n \geq 0 \}$	7	CO5
8	Draw a transition diagram for a Turing machine accepting the following language. $\{ x \in \{ a, b, c \}^* \mid n_a(x) = n_b(x) = n_c(x) \}$	7	CO5
9	Draw the TM to copy string and delete a symbol.	7	CO5
10	Draw Turing Machine(TM) which recognizes words of the form $\{ a^n b^n c^n \mid n \geq 1 \}$	7	CO5
11	Design a Turing machine for deleting nth symbol from a string w from the alphabet $\Sigma = \{0, 1\}$.	7	CO5
12	Give definition of Turing Machine. What do you mean by an instantaneous description of a Turing Machine?	4	CO5
13	Define a Turing Machine. Design a Turing machine for deleting nth symbol from a string w from the alphabet $\Sigma =\{0,1\}$.	7	CO5

14	Define Turing Machine and draw a TM to accept {a,b}*{aba}{a,b}*	7	CO5
15	Design a Turing machine to reverse the string over alphabet {0, 1}	7	CO5
16	Compare and contrast push down automata and Turing machine.	4	CO5
17	Enlist limitations of Turing machines.	4	CO5
18	Design a Turing machine which accepts the language consisting string which contain aba as a substring over alphabets {a, b}	7	CO5
19	Draw a Turing Machine that accepts the language $\{a^n b^n a^n \mid n \geq 0\}$ over {a,b}*. Also trace the TM on input string aaabbbaaa.	7	CO5
20	Draw a Turing Machine that accepts the language {xx | x ∈{a,b}*}. Also trace the TM on input string aa.	7	CO5
21	What is Turing Machine? Write advantages of TM over FSM.	7	CO5

Course Coordinator **Head of Department**